

Black-Box Domain Adaptation for Object Detection with Retention-Driven Knowledge Compression

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Abstract

Black-Box Domain Adaptation (BBDA) is a highly practical yet challenging strategy that enables the deployment of pre-trained detectors to new unlabeled target domains without accessing source data or models. Compared to previous domain adaptation studies, BBDA not only provides stronger data privacy protection but also offers greater portability. Despite growing interest, existing BBDA strategies remain difficult to apply directly to object detection, as most prior works focus on classification and segmentation tasks that do not involve bounding box localization and rely on different learning mechanisms. In this paper, inspired by lifelong learning, we propose Retention-Driven Knowledge Compression (RDKC), which applies a brain-inspired learning process to BBDA for object detection. Specifically, RDKC consists of two key components: Memory Retention (MR) and Scene Compression (SC). MR is designed specifically for object detection under the BBDA setting, where it performs memorized contrastive learning on partitioned regions to better utilize informative cues from reliable areas while filtering out potential noise from noise prediction labels. SC introduces a contrastive mechanism between near and far-view regions, which enables the model to better learn from far-view regions under the guidance of near-view cues. Experimental results demonstrate that under the BBDA setting, RDKC outperforms previous SOTA methods across all evaluated benchmarks, achieving superior performance improvements.

1. Introduction

To mitigate domain shift in deep neural network training [17], traditional unsupervised domain adaptation (UDA) methods [20, 24, 39, 40, 62] adapt models trained on fully labeled source domains to unlabeled target domains, thereby reducing the burden of data collection and annota-

tion. However, in privacy-sensitive scenarios such as personal medical records, access to source data is often restricted by privacy-preserving policies, limiting the practical deployment of UDA techniques. To address this limitation, source-free domain adaptation (SFDA) methods [16, 21, 31, 50, 55] have recently emerged, assuming access only to unlabeled target data and a pre-trained source model during adaptation. While SFDA reduces the risk of privacy breaches by using the model instead of raw source data, prior work [33] has shown that certain generative techniques [11, 13] can potentially reconstruct source data from the source model. Compared to other UDA settings, black-box domain adaptation (BBDA) offers stronger privacy guarantees and greater deployment flexibility. By relying solely on unlabeled target data and a black-box predictor trained on the source domain, e.g., a cloud API service [33], BBDA avoids privacy and security risks associated with source data or model exposure. As illustrated in Figure 1, the BBDA process and its advantages are detailed.

Recent visual studies [32–34, 52–54, 57, 59] on BBDA have primarily focused on image classification tasks. BBDA for object detection is notably more challenging than for classification tasks. While classification involves a global-to-category mapping, object detection requires joint learning of local features for both bounding box localization and category prediction. However, existing BBDA learning strategies lack optimization for localization, which limits their applicability to object detection tasks. Moreover, these methods typically rely on adaptive label smoothing [33] to store global features of the entire image and perform cumulative predictions, which is inconsistent with the dynamic and region-specific prediction mechanism of object detection.

In this paper, we propose a novel method *Retention-Driven Knowledge Compression* (RDKC), which provides a versatile solution for black-box domain adaptive object detection. RDKC is the first to address the BBDA task through lifelong learning in biological behavioral science [27, 37, 38, 49], which models human memory as a dy-

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Need	UDA	SFDA	BBDA
Labeled source data	✓	✗	✗
Source model	✓	✓	✗
Cloud API Prediction	✗	✗	✓
Unlabeled target data	✓	✓	✓
Ability			
Architectural flexibility	Hard	Medium	Easy
Privacy risk	High	Medium	Low

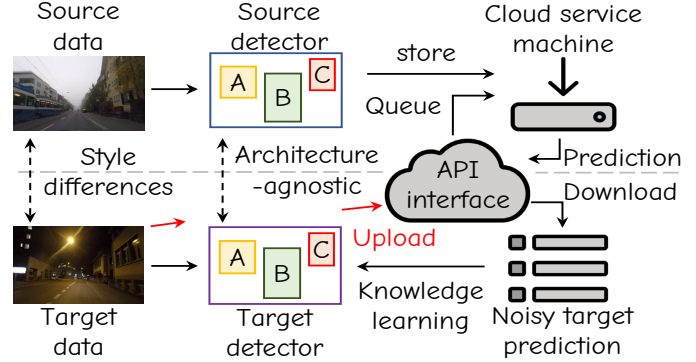


Figure 1. Comparison of DA settings. **(Left)** Compared with UDA and SFDA, BBDA provides better data privacy protection with more flexible portability. **(Right)** In BBDA, the cloud server only needs to store a source model pre-trained on source data and provides an API interface to users. Users upload target domain data with style differences to the API, which dispatches requests to the cloud server queue for prediction. The predicted batches are then returned to users for download. These noisy target predictions can be used for downstream knowledge learning. This process offers three key advantages: the API-based querying ensures data security, the batch scheduling improves efficiency, and the architectural independence between the source and target detectors enables more flexible deployment.

namic and adaptive system rather than a static repository. Human memory inherently balances three key processes to enable lifelong knowledge acquisition: 1) active forgetting, which identifies low-value information and reallocates cognitive resources for new learning; 2) selective stabilization, which prioritizes valuable information; and 3) collaborative integration across specialized modules, in which the outputs from high-robustness and low-robustness systems are jointly integrated to support cross-context learning. Specifically, RDKC comprises two main components: *Memory Retention* (MR) and *Scene Compression* (SC). Motivated by the observation of local image regions and the limitations of prior approaches, MR is a specific design for category optimization in object detection under BBDA settings, which emulates the active forgetting mechanism by reallocating contrastive learning across specific regions during training. This allows the model to suppress potential noise from noise prediction labels while making better use of informative cues from high-reliable regions. To enhance the fine-grained perception of bounding boxes for targets, we propose a contrastive module SC, which emulates the selective stabilization mechanism. In SC, we introduce a scene discrimination that adaptively compresses weights, enabling the model to learn far-view representations guided by highly reliable near-view cues. By conducting contrastive analysis of feature disparities between near-view and far-view object regions, the SC module enhances the model’s localization capability. Meanwhile, we introduce a teacher-student model [45] as the collaborative mechanism across specialized modules, integrating it into the contrastive learning process to prevent catastrophic forgetting. To better illustrate the applicability of our method to BBDA, we present a detailed comparison in Figure 2.

Our contributions can be summarized as follows:

- We identify critical limitations in existing domain adapta-

tion methods and tackle them by proposing a novel framework, named RDKC, inspired by biological behavioral science, which provides a versatile solution for black-box domain adaptive object detection.

- RDKC consists of *Memory Retention* (MR) and *Scene Compression* (SC). MR is specifically designed for object detection under BBDA to suppress noise from noise prediction labels and exploit informative cues from high-confidence areas. SC employs differential contrastive analysis between near- and far-view regions to enhance the fine-grained perception of bounding boxes.
- To evaluate the effectiveness of RDKC, we conducted extensive experiments and achieved SOTA performance across all benchmarks. Cross-method experiments further verified its versatility, while ablation studies revealed the contributions of individual modules and provided an in-depth analysis of their intrinsic correlations.

2. Related Works

2.1. UDA for Object Detection

Unsupervised domain adaptation (UDA) aims to transfer knowledge from a labeled source domain to a related but unlabeled target domain. It has attracted growing attention across various deep learning tasks, such as image classification [20, 39, 51], semantic segmentation [18, 19, 28, 62], time series forecasting [1, 24, 35, 41], and object detection [3, 26, 40, 42, 56]. In object detection, UDA is often formulated as unsupervised domain adaptive object detection (UDAOD). Despite previous works have made certain contributions, UDAOD requires access to both source and target data during training, which is impractical under privacy-preserving constraints that restrict the use of source data.

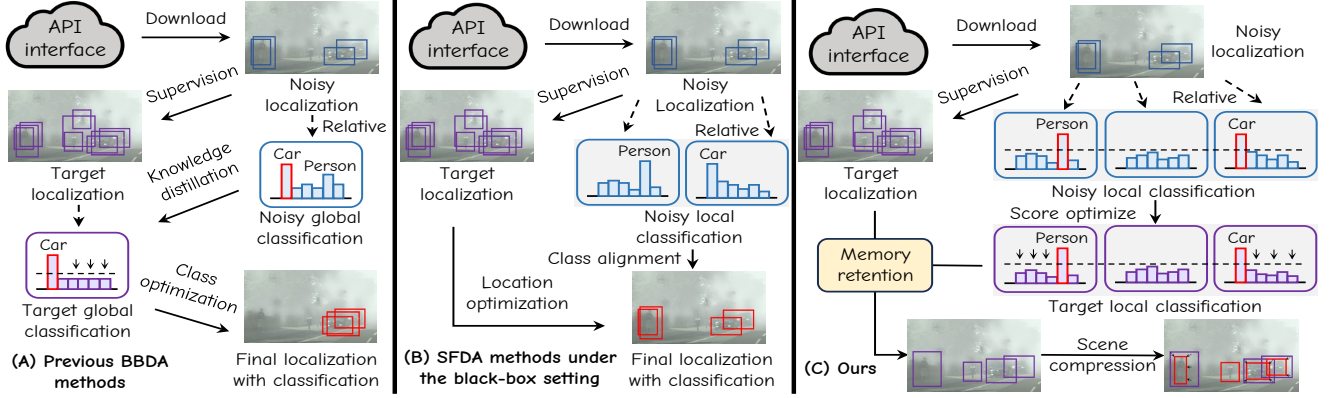


Figure 2. Illustration of the differences among previous BBDA methods, SFDA methods under the black-box setting, and our proposed RDKC. (A) Previous BBDA methods primarily focus on optimizing global image-level classification by suppressing negative predictions to better exploit reliable information. However, they fail to address the core challenges of detection tasks, such as object localization and local classification optimization. (B) Unlike source-free settings, SFDA methods under black-box constraints suffer from significant performance degradation. This is mainly due to the static nature of the downloaded predictions, which remain unchanged during the entire adaptation process. As a result, even advanced source-free optimization techniques are fundamentally limited, leading to only marginal improvements over the initial fixed outputs. (C) To address these limitations, we propose RDKC: MR suppresses noise using high-confidence regions while preserving blurred areas to balance noisy label learning and knowledge retention, and SC enhances bounding box perception through contrastive analysis.

2.2. SFDA for Object Detection

In comparison to UDA, source-free domain adaptation (SFDA) offers enhanced privacy protection by eliminating the need to access any source data. Recent advances in SFDA have led to the development of various methods [16, 21, 55] that require only unlabeled target data and a pre-trained source domain model during training. In object detection, SFDA is typically formulated as source-free domain adaptive object detection (SFOD). Existing SFOD methods mainly follow three technical paradigms: pseudo-label refinement [4, 30, 61], knowledge distillation [9, 29, 36], and domain alignment [5, 47, 55]. While SFOD has partially addressed privacy concerns, recent studies [33] reveal that exposing details of source-free model training still pose significant risks, primarily due to reverse generation techniques such as those proposed in [11, 13]. Detailed analyses of related works in UDAOD and SFOD are presented in Supplementary Material A.

2.3. BBDA for Object Detection

Recent advancements in generative adversarial networks (GANs) have significantly increased the sophistication of GAN-based white-box attacks, which now require access to source models to reconstruct original training data [33]. In contrast, BBDA eliminates the need for source data or model access, offering stronger privacy protection than other domain adaptation settings and reducing the risk of data breaches. Early work such as LNL-KL [57] explores noisy label learning through soft label utilization. More recent approaches, like DINE [33], adopt a two-stage process: first distilling knowledge to align source and target

classes, then fine-tuning the distilled model to fit the target distribution by leveraging reliable knowledge to cluster uncertain samples. Building on self-distillation, subsequent methods [52, 54] further divide the target domain into high- and low-reliability subdomains to reduce distribution gaps. BiMem [58] introduces information discrimination to prioritize informative samples over irrelevant ones during training. RFC [59] extends earlier works [33, 54] by incorporating neighborhood clustering to mitigate the forgetting of minority classes. Additionally, AEM [53] pioneers the use of multi-modal models as external prompts in BBDA, leveraging their rich semantic representations to enhance target domain feature alignment. However, existing studies remain primarily focused on classification tasks. In black-box domain adaptive object detection (BBOD), only BiMem has been proposed, which is derived from the classification method DINE. While classification centers on a global-to-category mapping, object detection requires joint learning of local features for both bounding box localization and category prediction. Current BBDA methods still lack well-established solutions for object detection tasks. To clarify these limitations, we provide a detailed analysis in Figure 2.

3. Proposed Method

3.1. Preliminaries

Statement. We begin by defining the unlabeled target domain $D_t = \{(x_i)\}_{i=1}^{N_t}$, where the N_t samples span K distinct classes. In the BBDA setting, the target dataset D_t is uploaded to a black-box predictor via a cloud API ser-

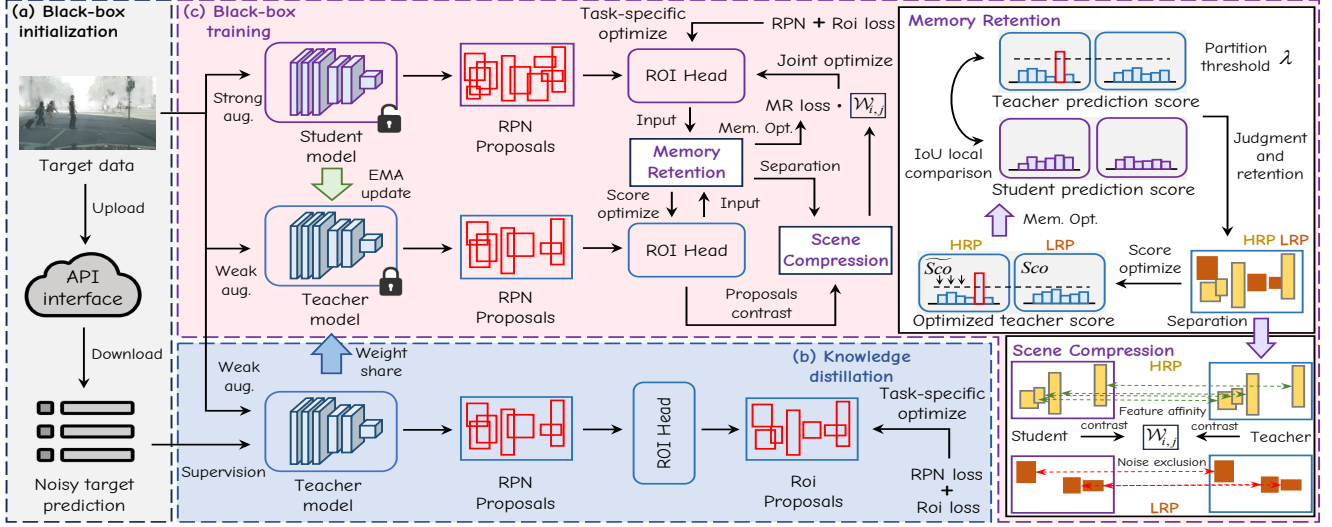


Figure 3. Overview of the whole training of RDKC. According to the BBOD, (a) target data can only be uploaded to the cloud API, and their corresponding noisy hard predictions must be downloaded without accessing the source model; (b) Subsequently, RDKC employs a simple knowledge distillation to enable the teacher model to learn target domain knowledge under the supervision of noisy hard predictions; (c) Based on the Mean-Teacher framework, RDKC freezes the teacher model’s parameters and initializes the student model by sharing the teacher’s weights. The student model is trained via our joint optimization, and the teacher is updated through EMA. Our joint optimization comprises two key components: Memory Retention, a BBOD-specific loss function designed to retain informative knowledge under noisy supervision; and Scene Compression, a region-aware contrastive analysis that enhances fine-grained perception of bounding boxes.

vice, and the corresponding noisy hard predictions P_s are retrieved from the API. This predictor hosts a source model trained on a labeled source domain D_s , which contains N_s labeled samples. Unlike low-security DA settings, BBOD constrains the adaptation process to the target model parameters Θ , which are trained without any access to source data or source models, relying solely on the noisy predictions from D_t . This setting reflects realistic deployment scenarios but poses significant challenges, as both the source data and the source-trained model remain entirely inaccessible.

Task-specific Loss and Mean-teacher Framework. Most domain adaptive object detection methods adopt the Mean-teacher self-training framework [45], where the teacher model generates pseudo labels from weakly augmented target data x_i^w to supervise the student detector’s learning on strongly augmented counterparts x_i^s . Following prior works [9, 36, 47], reliable pseudo labels are selected based on a confidence threshold. The pseudo-label supervision loss for the student detector is formulated as:

$$\mathcal{L}_{task} = \mathcal{L}_{cls}^{rpn}(x_i^s, \hat{y}_i) + \mathcal{L}_{reg}^{rpn}(x_i^s, \hat{y}_i) + \mathcal{L}_{cls}^{roi}(x_i^s, \hat{y}_i) + \mathcal{L}_{reg}^{roi}(x_i^s, \hat{y}_i), \quad (1)$$

where $\hat{y}_i$ represent pseudo labels of x_i^w , obtained from the teacher detector’s proposals via post-processing steps including score filtering (e.g., Non-Maximum Suppression) and confidence thresholding; $\mathcal{L}_{cls}^{rpn}$, $\mathcal{L}_{reg}^{rpn}$, $\mathcal{L}_{cls}^{roi}$, and $\mathcal{L}_{reg}^{roi}$ are the standard detection losses of Faster-RCNN [3]. $\mathcal{L}_{cls}^{rpn}$ and $\mathcal{L}_{reg}^{rpn}$ denote the classification and regression losses of

the Region Proposal Network (RPN) for candidate proposals, respectively, while $\mathcal{L}_{cls}^{roi}$ and $\mathcal{L}_{reg}^{roi}$ correspond to the classification and regression losses of the Region of Interest (ROI) head.

The teacher model parameters Θ_{tea} are updated via the Exponential Moving Average (EMA) of the optimized student model parameters Θ_{stu} . The overall Mean-teacher framework is formally updated as follows:

$$\Theta_{stu} = \Theta_{stu} + \alpha \frac{\partial(\mathcal{L}_{task})}{\partial(\Theta_{stu})}, \quad (2)$$

$$\Theta_{tea} = \beta \Theta_{tea} + (1 - \beta) \Theta_{stu}, \quad (3)$$

where hyperparameters α and β represent the student learning rate and the teacher EMA decay rate, respectively.

3.2. Lifelong Learning-inspired RDKC Framework

In lifelong learning [27, 38], human memory functions as a dynamic adaptive system that continuously balances three interdependent processes to sustain lifelong knowledge acquisition: (1) active forgetting, which selectively identifies and removes low-value or outdated information through molecular and neural mechanisms, thereby reallocating cognitive resources to facilitate new learning; (2) selective stabilization, which preserves high-value information by strengthening the associated neural pathways, ensuring the retention of critical knowledge and frequently used skills; and (3) collaborative integration across specialized modules, where distinct memory subsystems operate as

parallel learners, and the outputs from high-robustness and low-robustness systems are synergistically integrated to enhance cross-context generalization and adaptive reasoning.

Although the mean-teacher framework [45] facilitates the distillation of target-domain knowledge into the model in BBDA, it remains limited in its ability to learn target features without guidance from source knowledge. To address this limitation, inspired by lifelong learning, RDKC is developed based on two key principles: active forgetting and stability preservation. It introduces Memory Retention and Scene Compression to actively regulate the balance between memory preservation and forgetting. Meanwhile, RDKC integrates the multi-model collaborative optimization from the mean-teacher framework, where the teacher model guides the student model as the foundation for distributional compatibility. This integration within the detection frameworks enhances cross-context generalization and adaptive reasoning, which further enhances target feature discrimination. The RDKC training process is shown in Figure 3. *Detailed analyses of the relationship between lifelong learning and RDKC is presented in Supplementary Material B.*

3.3. Memory Retention

In the BBDA setting, source data and source-trained model are inaccessible, and the downloaded noisy target labels remain static throughout the adaptation process. How to effectively select and learn from reliable knowledge embedded in hard predictions while mitigating the negative effects of noisy supervision remains a core challenge in black-box tasks.

To solve this problem, we propose Memory Retention (MR), a mechanism inspired by active forgetting that distinguishes informative regions during training and reallocates scoring to suppress noise-related information. It first conducts knowledge distillation with noisy hard predictions P_s , and uses $\mathcal{L}_{task}$ to perform alignment learning on the teacher model parameters Θ_{tea} to extract the necessary target domain knowledge. The teacher detector updating formula can be expressed as:

$$\Theta_{tea} = \Theta_{tea} + \alpha \frac{\partial(\mathcal{L}_{task})}{\partial(\Theta_{tea})}. \quad (4)$$

Then, we follow the EMA update to lock the teacher detector and start the black-box training. In the black-box training, we first employ the teacher detector to perform a coarse-grained assessment of its predicted proposals and divide them into different types accordingly. The proposal type division formula is denoted as:

$$Type_{i,j} = \begin{cases} HRP_{i,j}, & \text{if } Top1(Sco_{i,j}^{tea}) < \lambda \text{ and} \\ & IoU(Pro_{i,j}^{stu}, Pro_{i,j}^{tea}) < mIoU(Pro_i^{stu}, Pro_i^{tea}), \\ LRP_{i,j}, & \text{otherwise} \end{cases} \quad (5)$$

where $HRP_{i,j}$ and $LRP_{i,j}$ indicate whether the j -th proposal in the i -th target sample is a high-reliability type proposal or a low-reliability type proposal; $Top1$ defines the largest value in the corresponding tensor; $Sco_{i,j}^{tea}$ defines the softmax predicted score for each class in the proposal box; λ is a hyperparameter that determines the partition threshold; IoU represents the Intersection over Union; $mIoU$ represents the mean Intersection over Union, which compares each proposal box and takes the average; $Pro_{i,j}^{stu}$ and $Pro_{i,j}^{tea}$ correspond to the j -th proposal in the i -th target data, predicted by the student and teacher predictors, respectively; and $Pro_i^{stu} = \sum_{k=1}^n Pro_{i,k}^{stu}$, $Pro_i^{tea} = \sum_{k=1}^n Pro_{i,k}^{tea}$, where n represents the number of proposal boxes of the teacher predictor. During the proposal-type division phase, we retain the type of each proposal in a memory bank to facilitate efficient retrieval and comparison during subsequent optimization.

Next, we perform memorized learning for different types of proposal boxes. For HRP -type proposal boxes, MR retains the highest prediction score and proportionally suppresses the others, aiming to leverage reliable knowledge to suppress the noise information influence. For LRP -type proposal boxes, MR retains the whole predicted scores, aiming to prevent catastrophic forgetting during model training. The prediction score optimization for HRP -type proposals under memory retention can be expressed as:

$$\widetilde{Sco}_{i,j}^{tea} = \begin{cases} Sco_{i,j,k}^{tea}, & k \in \mathcal{T}_{index}^K(Sco_{i,j}^{tea}) \\ \frac{\sum_{l=1}^K (1 - Sco_{i,j,l}^{tea}) \cdot Sco_{i,j,k}^{tea}}{K - 1}, & k \in otherwise \end{cases}, \quad (6)$$

where $\mathcal{T}_{index}^K(\cdot)$ denotes the index corresponding to the maximum predicted value among the K classes. Meanwhile, for LRP -type proposals, the optimized prediction scores remain unchanged. The total optimization formula for memory retention can be expressed as:

$$\begin{cases} Sco_{i,j}^{tea} = \alpha \widetilde{Sco}_{i,j}^{tea} + (1 - \alpha) Sco_{i,j}^{tea*}, Type_{i,j} = HRP_{i,j}, \\ Sco_{i,j}^{tea} = \alpha Sco_{i,j}^{tea} + (1 - \alpha) Sco_{i,j}^{tea*}, Type_{i,j} = LRP_{i,j}, \end{cases} \quad (7)$$

where $Sco_{i,j}^{tea*}$ represent the prediction score from the previous training epoch, which is used to prevent catastrophic forgetting in memory; α is set to 0.6 following BBDA methods [33, 52], representing the static coefficient to stabilize the updating.

The optimized prediction scores of our proposed MR outperform the original scores for the following reasons: (1)

Instead of applying uniform regularization across all classes as in label smoothing [44], MR focuses on fine-grained processing of different data types and reallocates noisy information (*i.e.*, the logits of non-maximal classes), which suppresses redundant information while mitigating catastrophic forgetting; (2) Inspired by self-weighted pseudo labeling [23], MR does not rely solely on noisy labels but determines confidence through joint learning and contrastive mechanisms between student and teacher models; (3) Unlike classification-driven methods [33, 52, 54], MR employs a local image partitioning strategy that balances noisy label learning and knowledge retention, enhancing learning efficiency while preventing overfitting.

To specifically optimize for the BBDA setting, MR applies a selective learning optimization loss to the student model. The memory optimization loss $\mathcal{L}_{MR}$ can be formulated as:

$$\mathcal{L}_{MR} = \frac{1}{n^{HRP}} \sum_{j=1}^{n^{HRP}} D_{KL}(Sco_{i,j}^{stu} || \widetilde{Sco}_{i,j}^{tea}) + \frac{1}{n^{LRP}} \sum_{j=1}^{n^{LRP}} D_{KL}(Sco_{i,j}^{stu} || Sco_{i,j}^{tea}) \quad (8)$$

where $D_{KL}(\cdot)$ is the Kullback-Leibler divergence; n^{HRP} represents the number of *HRP*-type proposal boxes; and n^{LRP} represents the number of *LRP*-type proposal boxes.

3.4. Scene Compression

In previous works [14, 43], they were observed that there is a positive correlation between the feature distance of student-teacher predictions in the same region and the region’s IoU with the ground-truth bounding box, *i.e.*, pseudo labels with high confidence often have higher IoU. Therefore, we design a contrastive module Scene Compression (SC), which performs contrastive learning between the predictions of the student and teacher models to adaptively learn features from regions with varying confidence levels and generate compression weights. SC simulates the selective stabilization mechanism by dynamically identifying high-value and low-value knowledge via weight discrimination, and uses the former to guide the learning of the latter, maintaining stability during the learning process. The scene compression weight $\mathcal{W}_{i,j}$ can be expressed as:

$$\mathcal{W}_{i,j} = \begin{cases} \exp(-\log[\cos(Sco_{i,j}^{stu}, \widetilde{Sco}_{i,j}^{tea})]), Type_{i,j} = HRP_{i,j} \\ \exp(-\log[1-\cos(Sco_{i,j}^{stu}, Sco_{i,j}^{tea})]), Type_{i,j} = LRP_{i,j} \end{cases} \quad (9)$$

where $\cos(\cdot)$ is the operation of calculating cosine similarity. The scene compression weight $\mathcal{W}_{i,j}$ focuses on high-reliability bounding boxes where student and teacher models show significant prediction discrepancies, using the teacher model as an “anchor” to force the student model to align with it and compress features. For low-reliability

Table 1. Quantitative results (mAP) under the weather adaptation scenario *Cityscapes* $\rightarrow$ *Foggy-Cityscapes*. The setting of **U**, **SF**, and **BB** corresponds to UDA, SFDA, and BBDA, respectively. $\dagger$ denotes methods originally designed for other settings but adapted to the BBDA scenario. **MR** means **RDKC (w/o SC)**. The top-performing BBDA methods are highlighted in **bold**.

Method	Setting	Person	Rider	Car	Truck	Bus	Train	Motor	Bicycle	mAP
Source-only	—	29.3	34.1	35.8	15.4	26.0	9.09	22.4	29.7	25.2
DA-Faster	U	25.0	31.0	40.5	22.1	35.3	20.2	20.0	27.1	27.6
MeGA-CDA	U	37.7	49.0	52.4	25.4	49.2	46.9	34.5	39.0	41.8
UnbiasedDA	U	33.8	47.3	49.8	30.0	48.2	42.1	33.0	37.3	40.4
TIA	U	31.1	49.7	46.3	34.8	48.6	37.7	38.1	52.1	42.3
PT	U	40.2	48.8	59.7	30.7	51.8	30.6	35.4	44.5	42.7
MT	SF	37.4	43.0	45.0	27.2	37.2	25.1	28.2	38.2	34.3
SFOD	SF	32.6	40.4	44.0	21.7	34.3	11.8	25.3	34.5	30.6
A ² SFOD	SF	32.3	44.1	44.6	28.1	34.3	29.0	31.8	38.9	35.4
LPU	SF	39.0	50.3	55.4	24.0	46.0	21.2	30.3	44.2	38.8
IRG	SF	37.4	45.2	51.9	24.4	39.6	25.2	31.5	41.6	37.1
PTVM	SF	38.4	47.1	52.7	24.3	44.6	36.3	30.2	40.1	39.5
LPLD	SF	39.7	49.1	56.6	29.6	46.3	26.4	36.1	43.6	40.9
DINE	BB	41.4	40.9	55.0	21.8	38.5	25.7	28.2	40.4	36.5
BiMem	BB	42.2	42.5	56.9	23.4	39.7	28.5	32.4	41.3	38.4
SEAL	BB	43.1	45.0	55.4	20.8	33.5	8.5	26.2	41.4	34.3
IRG [†]	BB	35.9	44.5	44.9	22.1	36.2	22.4	21.6	37.4	33.1
LPLD [†]	BB	34.7	42.6	45.6	19.6	35.7	20.7	22.5	39.5	32.6
IRG [†] +MR	BB	42.7	44.2	54.0	22.9	41.8	25.6	33.1	42.1	38.3
LPLD [†] +MR	BB	43.9	47.6	57.8	23.7	43.0	26.5	32.3	44.3	39.9
RDKC (Ours)	BB	45.4	45.9	56.1	24.3	44.7	29.1	33.2	46.9	40.7
w/o SC	BB	41.9	46.1	55.3	23.8	36.4	24.3	31.2	42.0	37.6

bounding boxes, $\mathcal{W}_{i,j}$ suppresses the learning of highly discrepant predictions and mitigates noise interference.

According to previous work [48], class-dominated bounding boxes tend to be more reliable in near-view scenes, while their confidence notably declines in far-view scenarios. By leveraging the cosine similarity between the student and teacher models, the SC module guides the memory optimization loss $\mathcal{L}_{MR}$ to prioritize the learning of high-confidence class-dominated bounding boxes over background regions. This enables the student model to learn far-view representations under the guidance of near-view visual cues. To realize the intended effect of SC, we combine the memory optimization loss $\mathcal{L}_{MR}$ with the scene compression weight $\mathcal{W}_{i,j}$, which can be formulated as:

$$\mathcal{W}_i \mathcal{L}_{MR} = \frac{1}{n^{HRP}} \sum_{j=1}^{n^{HRP}} \mathcal{W}_{i,j} \cdot D_{KL}(Sco_{i,j}^{stu} || \widetilde{Sco}_{i,j}^{tea}) + \frac{1}{n^{LRP}} \sum_{j=1}^{n^{LRP}} \mathcal{W}_{i,j} \cdot D_{KL}(Sco_{i,j}^{stu} || Sco_{i,j}^{tea}) \quad (10)$$

3.5. Overall Joint Optimization Function

The overall joint optimization loss in RDKC for updating the student model can be formulated as:

$$\mathcal{L}_{total} = \mathcal{L}_{task} + \eta \cdot \mathcal{W}_i \mathcal{L}_{MR} \quad (11)$$

Table 2. Quantitative results (AP of Car) under the synthetic-to-real adaptation scenario *Sim10K* $\rightarrow$ *Cityscapes* and cross-camera adaptation scenario *KITTI* $\rightarrow$ *Cityscapes*.

Method	Setting	Sim10K $\rightarrow$ Cityscapes	KITTI $\rightarrow$ Cityscapes
		AP of Car	AP of Car
Source-only	—	32.0	33.9
DA-Faster	U	38.9	38.5
MeGA-CDA	U	44.8	43.0
UnbiasedDA	U	43.1	—
PT	U	55.1	60.2
MT	SF	39.7	41.2
SFOD	SF	39.7	41.2
A ² SFOD	SF	44.0	44.9
LPU	SF	47.3	48.4
IRG	SF	45.2	46.9
PTVM	SF	48.6	48.7
LPLD	SF	49.4	51.3
DINE	BB	37.9	39.8
BiMem	BB	37.2	40.1
SEAL	BB	38.1	40.0
IRG [†]	BB	38.6	40.9
LPLD [†]	BB	38.9	41.3
IRG [†] +MR	BB	44.7	47.5
LPLD [†] +MR	BB	46.8	48.1
RDKC (Ours)	BB	49.1	50.4
w/o SC	BB	43.9	47.1

Table 3. Quantitative results (mAP) under the realistic-to-artistic adaptation scenario *Pascal-VOC* $\rightarrow$ *Watercolor*.

Method	Setting	Bike	Bird	Car	Cat	Dog	Person	mAP
Source-only	—	68.8	46.8	37.2	32.7	21.3	60.7	44.6
DA-Faster	U	75.2	40.6	48.0	31.5	20.6	60.0	46.0
ADDA	U	79.9	49.5	39.5	35.3	29.4	65.1	49.8
WST	U	77.8	48.0	45.2	30.4	29.5	64.2	49.2
MT	SF	73.6	47.6	46.6	28.5	29.4	58.6	47.1
SFOD	SF	76.2	44.9	49.3	31.6	30.6	55.2	47.9
IRG	SF	75.9	52.5	50.8	30.8	38.7	69.2	53.0
LPLD	SF	86.0	51.8	49.6	32.9	40.0	70.8	55.2
DINE	BB	74.1	44.3	45.8	30.4	23.4	55.8	45.6
BiMem	BB	73.5	43.8	44.9	30.1	22.9	56.0	45.2
SEAL	BB	75.0	46.0	47.2	32.0	24.5	52.0	46.1
IRG [†]	BB	73.9	46.7	48.4	28.7	34.3	60.9	48.8
LPLD [†]	BB	75.4	48.8	48.8	29.6	37.9	63.4	50.7
IRG [†] +MR	BB	74.6	50.3	49.6	30.2	38.4	63.1	51.0
LPLD [†] +MR	BB	77.1	51.2	50.9	32.6	42.1	64.2	53.1
RDKC (Ours)	BB	76.5	52.1	52.4	35.2	38.8	66.9	53.7
w/o SC	BB	73.6	49.9	50.1	29.3	36.4	61.1	50.1

where η is a hyperparameter that controls the effect of our proposed approach RDKC. The training process of RDKC is shown in Algorithm 1 of Supplementary Material D.

4. Experiments

Datasets. The evaluation is conducted on six widely used domain adaptive object detection benchmarks, covering four domain shift scenarios: *Cityscapes* [6] $\rightarrow$ *Foggy-Cityscapes* [15], *Sim10k* [25] $\rightarrow$ *Cityscapes* [6], *KITTI* [12] $\rightarrow$ *Cityscapes* [6], and *Pascal-VOC* [10] $\rightarrow$ *Watercolor* [22]. Specific dataset details are presented in Supplementary Material C.

Implementation Details. Our implementation is built upon

PyTorch with all experiments conducted on an NVIDIA GeForce RTX4090 GPU. To make fair comparisons, following previous methods [47, 55], we adopt Faster R-CNN [3] as our default object detector, employing a ResNet-50 [17] backbone pre-trained on ImageNet [7]. The optimizer is stochastic gradient descent (SGD) configured with a learning rate α of 0.001, momentum of 0.9, and weight decay of 0.0001. The Exponential Moving Average (EMA) rate β for updating the teacher model is set to 0.8. The knowledge distillation stage is trained for 5 epochs, followed by 10 epochs in the black-box training stage. *More details are presented in Supplementary Material D.*

Comparison Methods. We evaluate the performance by comparing RDKC with several related methods across UDA, SFDA, and BBDA settings. For UDA, we conduct comparisons with DA-Faster [3], MeGA-CDA [46], UnbiasedDA [8], TIA [60], and PT [2]. For SFDA, we compare with MT [45], SFOD [30], A²SFOD [5], LPU [4], IRG [47], PTVM [9], and LPLD [55]. For BBDA, we compare our method with previous SOTA approaches, DINE [33], BiMem [58], and SEAL [52]. Note that BiMem does not provide a complete solution for object detection, while DINE and SEAL are re-implemented by us for fair comparison. For a more comprehensive comparison, we adapt the previous SFDA methods IRG and LPLD to the BBDA setting under the same experimental conditions.

Qualitative and Quantitative Results. As shown in Tables 1, 2, and 3, the results demonstrate that RDKC significantly outperforms previous SOTA methods across all four adaptation scenarios. Compared with the previous BBDA method DINE, RDKC achieves a +4.2 improvement in mean Average Precision (mAP) on *Cityscapes* $\rightarrow$ *Foggy-Cityscapes* and +8.1 on *Pascal-VOC* $\rightarrow$ *Watercolor*. Additionally, the average precision (AP) for the car category improves by +11.2 on *Sim10K* $\rightarrow$ *Cityscapes* and by +10.6 on *KITTI* $\rightarrow$ *Cityscapes*. Moreover, we construct cross-method experiments to validate that MR is specifically designed for the BBDA setting, effectively addressing the performance limitations of prior SFDA methods under this challenging setup. When integrated with previous SFDA methods IRG and LPLD, MR also exhibits remarkable effectiveness.

In Figure 4, we conduct qualitative experiments to compare our method RDKC with the baseline model, the source-only trained model, and the LPLD and SEAL methods. These results indicate that RDKC yields higher-quality predictions, outperforming the compared methods in both localization accuracy and semantic confidence. *Supplementary Material E presents additional quantitative experiments on more datasets, resource consumption comparisons, and qualitative studies demonstrating the architectural versatility of our framework.*

Parameter Sensitivity Analysis. As shown in Figure 5, we report the performance of models trained for 10 epochs

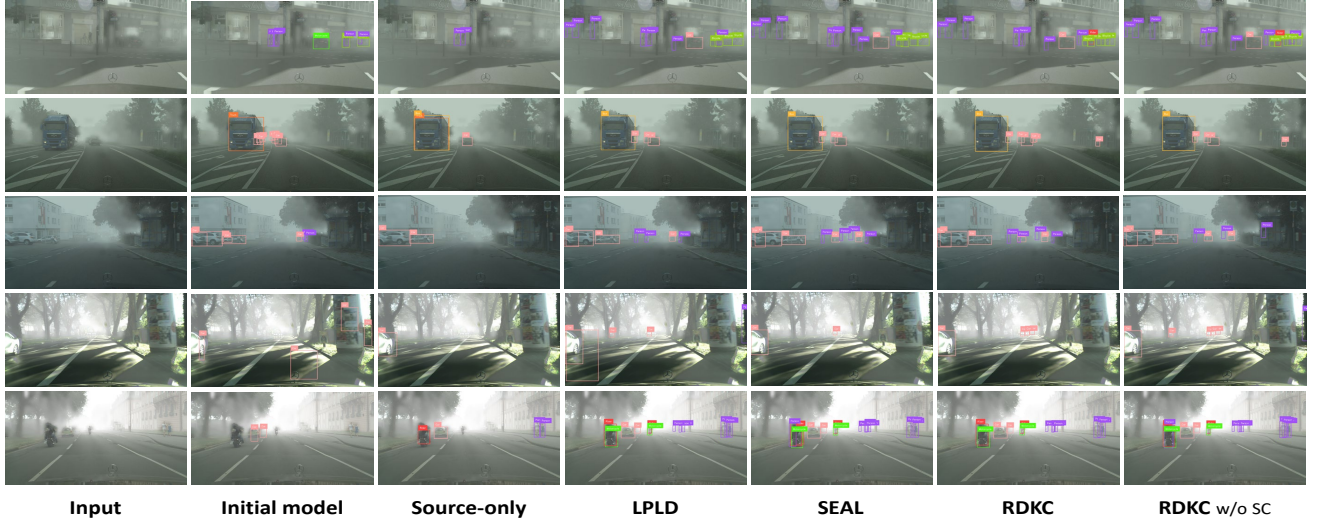
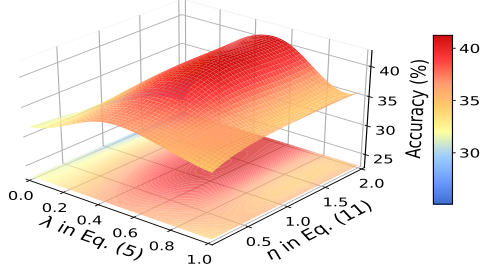


Figure 4. Qualitative results for the weather adaptation scenario *Cityscapes* $\rightarrow$ *Foggy-Cityscapes*. The bounding boxes indicate the predicted results. (Zooming in for a clearer view)



<i>City</i> $\rightarrow$ <i>Foggy</i>	$\lambda=0$	$\lambda=0.2$	$\lambda=0.4$	$\lambda=0.6$	$\lambda=0.8$	$\lambda=1.0$
$\eta=0.1$	31.7	33.8	36.4	36.5	35.2	34.1
$\eta=0.5$	31.2	34.5	37.7	38.0	36.6	35.6
$\eta=1.0$	30.1	35.6	39.6	40.5	36.4	36.1
$\eta=1.5$	28.6	35.7	40.0	40.7	35.9	36.4
$\eta=2.0$	25.1	35.2	40.1	40.4	36.2	35.9

Figure 5. Sensitivity analysis of the partition threshold λ and optimization factor η on *Cityscapes* $\rightarrow$ *Foggy-Cityscapes*.

on *Cityscapes* $\rightarrow$ *Foggy-Cityscapes* under varying values of λ and η , along with the corresponding 3D accuracy landscape. When $\lambda = 0$, all proposals are categorized as *LRP* type: MR becomes inactive, and SC uniformly suppresses the connections between each pair of teacher and student predictions, leading to the complete failure of our joint optimization loss. When $\lambda = 1$, all proposals are classified as *HRP* type: MR indiscriminately optimizes all prediction scores, while SC enforces strict alignment of the student model to the teacher model, thereby hindering the student from learning through their differences. The parameter η is an optimization factor that controls the effectiveness of our method, and choosing an appropriate value enables it to work more effectively.

Ablation Study. As shown in Tables 1, 2, and 3, we conduct quantitative ablations under different scenarios, where

w/o SC denotes RDKC without SC, implemented by fixing $W_{i,j}$ to 1. The results demonstrate that MR is a loss function specifically designed for BBDA and plays an essential role in such scenarios. When integrated with existing SFDA methods, MR effectively mitigates performance degradation caused by the static nature of noisy target labels. When MR and SC are combined, SC guides the memory optimization loss $\mathcal{L}_{MR}$ to focus on learning class-dominant bounding boxes with high confidence rather than background regions. In Figure 4, we observe that without SC, the model exhibits limited scene understanding: high-confidence class-dominant regions (e.g., the close-up target in the second row) show overlapping annotations, while low-confidence far-view objects are often missed. When our method is fully applied, RDKC continuously refines the learning of near-view regions while suppressing interference from distant areas, enabling the student model to learn far-view representations guided by near-view visual cues. As shown in Figure 4, even fine-grained objects in distant scenes can be accurately perceived.

5. Conclusion

In this work, we observe some weaknesses in existing BBDA methods and address them by proposing a novel method, RDKC. RDKC is a versatile solution for black-box domain adaptive object detection, consisting of two key components: *Memory Retention* (MR) and *Scene Compression* (SC). MR is specifically designed to optimize category-level predictions in BBDA scenarios, mitigating the performance degradation observed in previous methods lacking access to source models. SC is a contrastive module that enhances the fine-grained perception of bounding boxes. Experimental results show that RDKC consistently outperforms previous SOTA methods across all benchmarks.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China under Grant 62176162 and in part by Guangdong Basic and Applied Basic Research Foundation under Grant 2023A1515012875 and Grant 2022A1515140099.

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